PAN-DORSET GUIDELINE FOR THE USE OF FERRIC MALTOL

An option for treating iron deficiency anaemia in adults (not in pregnancy) who would otherwise be referred for intravenous iron

Introduction

This guideline covers specific use of ferric maltol for this condition and patient group. Please refer to other pathways for a full investigation of iron deficiency anaemia.

Iron deficiency anaemia (IDA) is the most common cause of anaemia, affecting 2-5% of adult men and post-menopausal women. The prevalence is increased in women of child-bearing age (14%), due to iron loss through menstruation.(1) Iron deficiency is associated with fatigue, impaired physical and cognitive function, headache, tachycardia, and dyspnea.(2)

Causes of IDA include:
- Poor dietary intake of iron – please assess patient's diet as a 1st line option for resolving iron deficiency
- Malabsorption of dietary iron (e.g. in Coeliac Disease)
- Chronic blood loss e.g. menstrual /GI blood loss
- Cancer

Anaemia is defined as haemoglobin (Hb) below normal range (<120g/L Females, <130g/L Males)

Iron deficiency anaemia (IDA) should be confirmed by iron studies prior to investigation; a full haematinics profile should be obtained. Low serum ferritin is the most useful marker, but low transferrin saturations can be a useful indicator of IDA if a false-normal ferritin is suspected (ferritin levels can be elevated by inflammatory processes and can mask iron deficiency).(1)

A anti-tissue transglutaminase (tTG) and IgA level should be also taken to assist screening for coeliac disease (found in 3-5% of cases)(1)

1. Initial investigation and 1st line treatment of confirmed IDA

Further investigation by secondary care of confirmed IDA should include urinalysis /urine microscopy, screening for coeliac disease and where appropriate endoscopy of upper and lower GI tract. (NB not warranted in young women unless clinical features of concern, as underlying GI pathology is uncommon in young women with IDA).(1)

Iron treatment can be started while awaiting investigations unless colonoscopy is imminent.(1)

PARENTERAL IRON should be used if oral iron is contraindicated or if correction of IDA is urgent (including if Hb is less than 100g/L)1). The IV route may also be preferable if ongoing significant bleeding, malabsorption, combination of iron deficiency and anaemia with significant active inflammation, or issues with oral route /compliance.(1)

Please refer to the UHD intravenous iron policy for information on the IV route.

NOTE: GPs may refer patients directly to the Treatment and Investigation Unit (TIU) at UHD for iron infusions.

1st line oral treatment

FERROUS SULPHATE 200mg, FERROUS FUMARATE 210mg or FERROUS GLUCONATE 300mg ONCE a day

- The ferrous sulphate and fumarate options will give the recommended 50-100mg elemental iron daily. Ferrous gluconate 300mg once a day is another option recommended by NICE CKS[2] (but contains 37mg of elemental iron).
- NOTE modified release preparations are not recommended – they do not enhance iron absorption or reduce side effects
- Co-administration with vitamin C is not recommended (no evidence of benefit).

- Iron preparations should ideally be taken on an empty stomach first thing in the morning (taking with meals can reduce bioavailability by up to 75%), even a small amount of milk in tea can reduce absorption.
- Drugs such as antacids and tetracyclines reduce absorption. Separate administration by at least 2 hours.
- Counsel the patient that they may experience adverse effects from iron supplements but these usually settle with time.[2]

2. Monitoring of efficacy of oral iron treatment in IDA

Haemoglobin levels should be checked after at least 4 weeks of treatment for a response to oral iron. Aim for Hb rise of $\geq$20g/L over 4 weeks.[2] Hb levels normalise with iron replacement in most cases of IDA.

Continue treatment for around 3 months after normalisation of Hb to ensure repletion of the marrow iron stores. Then monitor blood count every 6 months for 2-3 years to detect recurrent IDA (IDA can recur in a minority of patients). [2]

Failure to respond to oral iron has many causes including non- compliance, malabsorption, systemic disease, bone marrow pathology, haemolysis, continued bleeding and concurrent deficiency of vitamin B12 or folic acid.[1]

3. Second line options for oral treatment of IDA where 1st line option is not tolerated

Poor compliance due to adverse effects should be addressed where possible e.g. offer a laxative to people with constipation, offer reassurance to people with black stools.

The 1st line iron preparation chosen can be reduced to alternate day dosing if not tolerating daily dosing due to side effects (e.g. constipation, diarrhoea, nausea, GI discomfort).
The iron supplement could be taken with/after food which may increase tolerability, however this may decrease iron absorption.

An alternative first line iron salt (as above) may be trialled if the first preparation is not tolerated.

4. Third line treatment of IDA where two standard oral iron preparations are not tolerated

Intolerance to standard oral iron preparations may persist despite reduction to alternate day dosing or taking with food, at which point alternative options must be considered **– referral to secondary care for intravenous iron, or trial oral ferric maltol.** [2]

FERRIC MALTOL is a relatively new preparation, licensed for treatment of IDA of any cause. This ferric maltol iron complex has high bioavailability and is designed to be better tolerated than standard ferrous salts.[3]

Ferric maltol oral preparation has an increased cost in comparison to standard oral iron preparations, but has a favourable GI side effect profile. There is no evidence that ferric maltol is any more efficacious than conventional iron preparations, but it may be better tolerated. It should also be considered that although more expensive than traditional iron salts, ferric maltol is considerably less expensive than parenteral irons.[1]

Contraindications to ferric maltol include: Exacerbation of inflammatory bowel disease (IBD), haemochromatosis, IBD with Hb <95g/L, repeated blood transfusions, and iron overload syndromes.[4] Care is needed in intestinal stricture and diverticular disease.

In chronic kidney disease ferric maltol is not recommended if patients are on dialysis or having erythropoiesis - stimulating agents (ESAs) or a hypoxia-inducible factor, prolyl hydroxylase inhibitors (HIF-PHI) e.g., Roxadustat. These patients should still receive intravenous iron.

FERRIC MALTOL can be **trialled** if a patient is intolerant to **at least two** standard oral iron preparations (including at alternate day dosing), and has no contraindications. Further investigation is also then needed into the cause of the deficiency.

FERRIC MALTOL standard dose is 30mg twice a day (morning and evening) on an empty stomach, until iron stores are replenished (usually at least 12 weeks).
Refer to Appendix A for flow chart on use of Ferric Maltol.

Ferric maltol has a relatively low iron content (30mg elemental iron per tablet), so the dose must be two tablets daily (1 bd). This potential increased frequency should be taken into consideration if the patient had compliance issues with taking once daily standard iron preparations.

No dose adjustment of ferric maltol is required in elderly patients or in renal impairment.

Pharmacological interactions: [(5)]
Oral calcium and magnesium salts may reduce absorption of ferric maltol. Recommended to separate administration by at least 2 hours.
Absorption of the following medications may be reduced by ferric maltol. Recommended to separate administration by at least 2 hours:

- Bisphosphonates
- Ciprofloxacin
- Entacapone
- Levofloxacin
- Levodopa
- Levothyroxine
- Moxifloxacin
- Mycophenolate
- Norfloxacin
- Penicillamine
- Ofloxacin
- Tetracycline

Avoid the use of the following medications alongside ferric maltol:
- iv iron- may lead to hypotension and collapse
- Methyldopa- may antagonise hypotensive effect of methyldopa
- Chloramphenicol (delays clearance of plasma iron, impairing erythropoiesis)
- Dimercaprol- nephrotoxic

ADVERSE EFFECTS:
Most common side effects include abdominal pain (8%), flatulence (4%), constipation (4%), abdominal discomfort(2%)/distention(2%), diarrhoea (3%) of mild-moderate intensity.

MONITORING OF FERRIC MALTOL:
Assess tolerability of ferric maltol, and Hb response.
Ferric maltol is significantly more expensive in comparison to other oral iron preparations (comparative cost is approximately £1 per month for once daily ferrous sulphate vs £48 per month for twice daily ferric maltol)[(1)] , so it should be carefully monitored for efficacy and be used in line with this guidance document.

5. References

1. Snook J, Bhala N, Beales I et al (2021) British Society of Gastroenterology guidelines for the management of iron deficiency anaemia in adults. *Gut* 2021; 0:1-22
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4. BNF online accessed 15/5/22 at https://bnf.nice.org.uk/
5. emc (2021) SPC for Ferracru 30mg hard capsules. Accessed online at https://www.medicines.org.uk/emc/medicine/31722 Last updated Oct 2021.
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9. BMJ (2017);357:j2513 doi: 10.1136/bmj.j2513 (Published 2017 June 15) Interpreting Iron Studies
10. BSW Pathway for use of ferric maltol (Ferraccru) Feb 2022. -Adapted from original work by Notts APC.
11. Ballesteros, A (2020) UHD Guideline for the Use of Intravenous Iron (Cosmofer and Monofer) Version 3

APPENDIX A

Ferric Maltol (Feraccru®) Capsules for the treatment of iron deficiency anaemia in adults (NOT in pregnancy), for patients who are intolerant to conventional oral iron preparations and who would otherwise be referred for IV iron

Only for use following assessment of cause of iron deficiency anaemia (IDA) and where patient has not tolerated at least two conventional oral iron salts e.g. Ferrous sulphate/ferrous fumarate (see inclusion criteria below*)

Start ferric maltol 30mg BD for one month.

Repeat FBC and ferritin at 1/12 to assess if Hb and ferritin are improving or have normalised

Hb decreased

Refer to secondary care: Stop ferric maltol and consider IV iron

Hb and ferritin the same or increasing

Continue ferric maltol 30mg BD for a further 2/12 and repeat Hb and Ferritin after the full course of 3/12 to monitor response

Hb and ferritin normalised

Stop ferric maltol. Recheck Hb + ferritin after treatment gap of 3 months.

Insufficient rise in ferritin and Hb

At a dose of 30mg BD, normalisation of Hb and Ferritin would be expected in 3 months. GP can extend to maximum of 6 months for single course if required or refer for consideration of IV iron.

Ferric maltol inclusion criteria:*

FERRIC MALTOL can be **trialled** if a patient is intolerant to **at least two** standard oral iron preparations (including at alternate day dosing), and has no contraindications. Further investigation is also then needed into the cause of the deficiency.

1. Hb and ferritin failed to normalise after adequate trials of at least two conventional oral iron salts and patient has no exclusion criteria/contraindications.

2. Deemed intolerant of conventional oral iron salts following an individual patient review and side effects (e.g. constipation, diarrhoea, epigastric pain, faecal impaction, nausea, GI irritation, exacerbation of IBS) are unmanageable to the extent patient is no longer able to continue conventional oral iron despite dose reduction.

Ferric maltol exclusion criteria /contraindications:

- No previous trial of two oral iron salts
- Iron overload syndromes
- IBD patients with an active flare or Hb <9.5g/dL
- Repeat blood transfusions
- Haemochromatosis
- In chronic kidney disease ferric maltol is not recommended if patients are on dialysis or having erythropoiesis -stimulating agents (ESAs) or a hypoxia-inducible factor, prolyl hydroxylase inhibitors (HIF-PHI) e.g., Roxadustat. These patients should still receive intravenous iron.

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Intolerance to oral iron (from NICE CKS topic Anaemia – iron deficiency last revised Nov 21)

- Adverse effects are dose related and are directly related to the amount of iron absorbed (although the relationship between constipation or diarrhoea is less clear than for nausea and epigastric pain).

- The incidence of adverse effects is no greater with ferrous sulfate than with other iron salts.

- Adverse effects of oral iron supplements are a common cause of non-compliance with treatment — 10–20% of people are thought to discontinue iron supplements because of adverse effects.

<u>If adverse effects are troublesome, the following may help to minimize them:</u>

- Taking the iron supplement with or after food. Iron supplements are better tolerated when taken with or after food. However, this may decrease iron absorption by 40–66%.

- Reducing the dose frequency to alternate days

Repeating courses: Courses may be repeated as and when necessary but **should be stopped if patient switches to IV iron.**

Further information: Feraccru® should be swallowed whole and taken on an empty stomach. Feraccru® contains lactose and gelatin of bovine origin. For full prescribing info see Feraccru Summary of Product Characteristics.